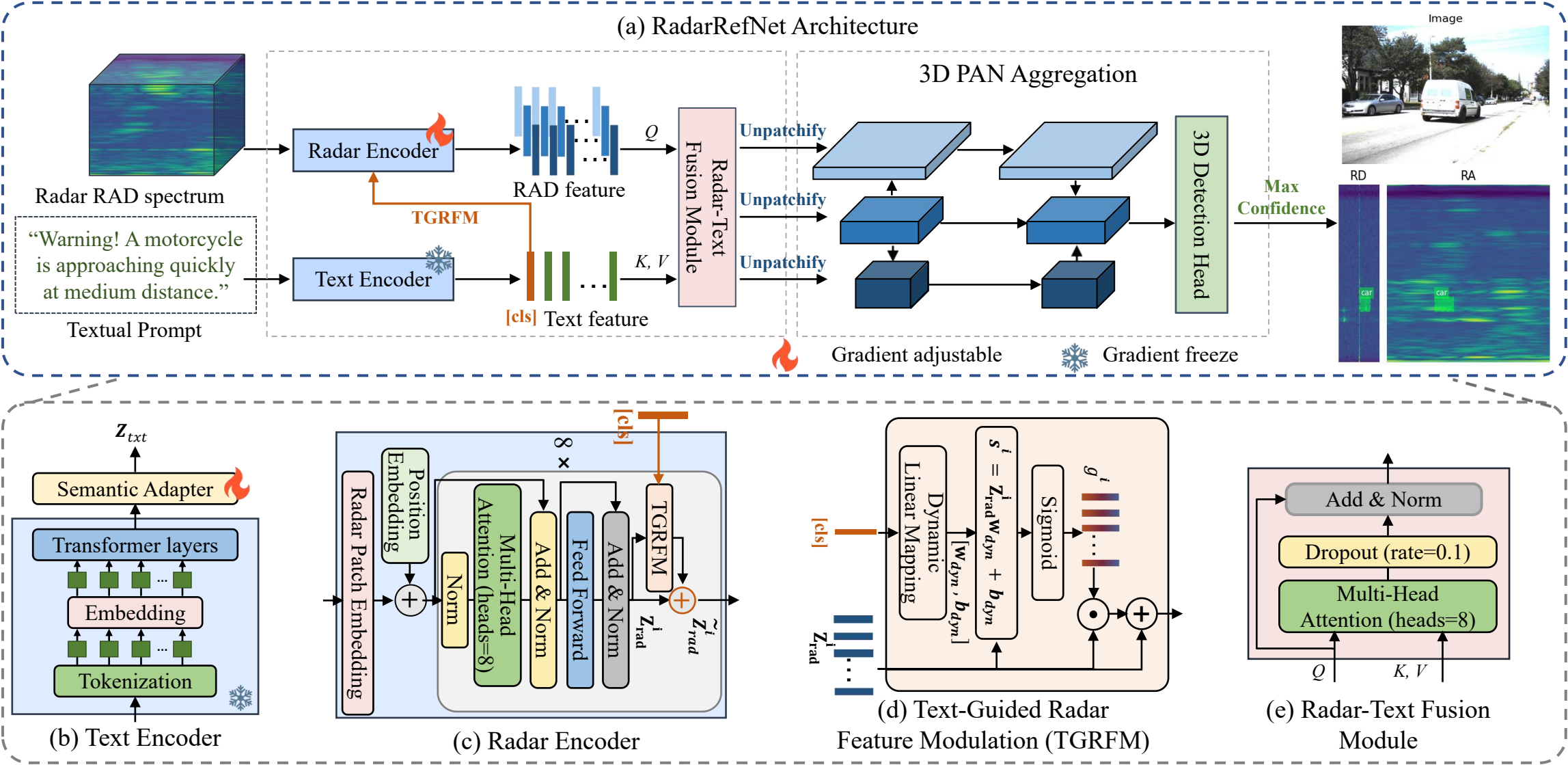

(a) RadarRefNet Architecture
Radar RAD spectrum
Radar Encoder
RAD feature
Q
TGRFM
"Warning! A motorcycle is approaching quickly at medium distance."
Textual Prompt
Text Encoder
[cls] Text feature
K, V
Radar-Text Fusion Module
Unpatchify
Unpatchify
Unpatchify
3D PAN Aggregation
3D Detection Head
Max Confidence
Image
RD
RA
car
car
Gradient adjustable
Gradient freeze
$z_{txt}$
Semantic Adapter
Transformer layers
Embedding
Tokenization
(b) Text Encoder
[cls]
8 ×
Position Embedding
Radar Patch Embedding
Norm
Multi-Head Attention (heads=8)
Add & Norm
Feed Forward
Add & Norm
TGRFM
$z^i_{rad}$
$\tilde{z}^i_{rad}$
(c) Radar Encoder
[cls]
$z^i_{rad}$
Dynamic Linear Mapping
$[w_{dyn}, b_{dyn}]$
$s^i = z^i_{rad} w_{dyn} + b_{dyn}$
Sigmoid
$g^i$
(d) Text-Guided Radar Feature Modulation (TGRFM)
Add & Norm
Dropout (rate=0.1)
Multi-Head Attention (heads=8)
Q
K, V
(e) Radar-Text Fusion Module